

John Silva.

MSC MECHANICAL ENGINEERING / TECHNICAL DIRECTOR & CO-OWNER,
CWT GROUP

I make difficult welds possible. Since 2012, turning materials that *shouldn't* weld cleanly into production parts that pass aerospace and automotive inspection — backed by **FEA, metallurgy and published research**, not guesswork.

- AEROSPACE
- AUTOMOTIVE · PPAP / PFMEA
- EV BATTERY · E-MOBILITY
- FORMULA 1
- MEDICAL
- OIL & GAS
- TOOLING



Tripled

Turnover, from the FY2017 low to FY2026, following the management restructure

INTERNAL SAGE LEDGER · NON-TRADING INCOME EXCLUDED

+10%

Average gain in ultimate tensile strength with ultrasonic assistance; +25% in the best trial

PEER-REVIEWED · METALS 2022

2012

The year I started in laser welding, and have worked in it continuously since

14 YEARS AND COUNTING

3

Management systems written and taken through to certification

AS9100D AT CARR'S WELDING PRODUCTS · ISO 9001 & ISO 14001 AT CWT

The engineer you call when the weld keeps cracking

Most of my work starts with a phone call about a part that will not behave. A weld that cracks on the fifth segment but not the first. An alloy the datasheet says is unweldable. A joint that has to survive 100,000 miles and a safety factor of three.

I graduated from Instituto Superior Técnico in Lisbon with an MSc in Mechanical Engineering, having first arrived at Carr's in February 2010 to run the welding trials for my thesis — on the floor of the very company that then hired me. Since 2012 I have worked my way from the shop floor to **Technical Director and co-owner of CWT Group** — a precision laser welding subcontractor serving aerospace, automotive, EV battery, Formula 1, medical, oil & gas and tool engineering. In 2025 I completed a management buyout of the business alongside my two fellow directors.

What I bring is the whole chain, not one link of it: assessing weldability at quote stage, calculating the weld depth and profile a design actually needs, developing the laser parameters to achieve it, and then building the quality system that proves it — every time, to ISO 9001. When the existing technology could not reach far enough, I went and improved the technology itself, and published the results.

AT A GLANCE	
ROLE	Technical Director & co-owner — CWT Group
BOARD	Shareholder since November 2018; statutory director since September 2021; co-owner from October 2025
BASED	Kettering, Northamptonshire, United Kingdom
EDUCATION	MSc & BEng Mechanical Engineering — Instituto Superior Técnico, Lisbon
SPECIALISM	Laser welding & marking, weld metallurgy, weld strength analysis
STANDARDS	ISO 9001:2015 and ISO 14001 at CWT; AS9100D written and certified at Carr's Welding Products

I am equally comfortable in front of a customer's engineering team, a conference audience, or a micrograph at 4000× magnification. The common thread is a diligent, methodical approach and a genuine enjoyment in explaining the *why* to whoever needs to hear it.

“The legacy I want to leave behind is a legacy of kindness and helpfulness — sharing what I have come to learn with people who will grow from it and become better engineers.”

PERSONAL MISSION

PUBLISHED	<i>Metals</i> 12(6):1041, 2022 · <i>The Laser User</i> #106
LANGUAGES	English (Cambridge CPE, C2) · Portuguese (native)
OFF THE CLOCK	Portuguese national triathlon champion, 2003. Since 2024, bladesmithing: heat-treating tool steel in a PID-controlled furnace at home, and a book for beginners in draft.

02 — CAPABILITIES

What I am usually brought in to do

Six areas where I take responsibility for the outcome — from the first weldability question on a drawing through to the certification pack that ships with the part.



Weldability & materials assessment

Evaluating whether a material and joint can be welded reliably — at quote stage, before anyone commits — and designing strategies around poor weldability rather than discovering it in production.

Alloy metallurgy

Hot & cold cracking

Dissimilar joints

Macro / micro analysis



Weld strength & design analysis

Calculating the weld depth and profile a design genuinely requires to hit its safety factor, and advising the customer at design stage — using SolidWorks, Cosmos and Ansys to prove it before cutting metal.

FEA — traction & torque

Safety factors

Joint redesign

Residual stress



Laser process development

Deriving the parameter set for every laser operation, guiding R&D programmes, assessing and procuring new technology, and training operators to run it. Including techniques that did not exist until we built them.

Disk & fibre lasers

Oscillating (wobble) welding

Ultrasonic-assisted

Parameter modelling



Quality systems & compliance

Writing, implementing and maintaining the management systems that let a subcontractor sell into aerospace and automotive — then running the audits, inspections and documentation that keep them credible. Fourteen years running the system; twice-yearly management reviews signed by all directors.

ISO 9001:2015

ISO 14001

AS9100D (CWP)

PFMEA / PPAP

FAIR / CoFC / IR



Laser marking & engraving

Designing and running laser marking since 2014 — programming, fixture design, depth and contrast control on steels, aluminium and tooling, plus operator training and process support.

TruTops Mark

Deep engraving

Traceability marking

CNC programming



Operations & commercial strategy

Reading a business through its own data. Building the KPIs, spotting where turnaround time and non-conformance are quietly killing margin, and rebuilding the plan around what the numbers actually say.

KPI design

Power BI · Excel / VBA

PERT · CPM · Gantt

Workflow redesign

Engineering tools I have built

Software written to the same standard as the quality system it serves.

2013 → NOW

Laser Settings Predictor — an Excel calculation tool I built in 2013 to derive starting laser parameters and predicted penetration depth from a job's material, thickness and target depth. In use at quote and setup stage ever since; now being ported to a standalone desktop application so it works without me in the room.

2026

WPS Database — desktop application for authoring and printing Welding Procedure Specifications to ISO 15609-4. Offline by design as a customer-confidentiality control, verified by a zero-network sweep of every screen. Six Architecture Decision Records, SHA-256 build identity, and a 247-check verification harness with mutation coverage. A data-entry fix caught a fivefold error in a vacuum-leak acceptance threshold on a real procedure before it shipped.

2026

KPI Dashboard — see Case 05.

Five problems, and what I did about them

Each one opens into the full engineering story — the context, the constraint, the method and the measured result. The micrographs and macros are from the actual samples.

Case 01

R&D

PEER-REVIEWED

ALUMINIUM AA6082

Making an “unweldable” aluminium alloy stronger with sound

AA6082 contains roughly 1% magnesium silicide, which makes it notoriously prone to hot cracking. The narrow parameter window that avoids cracking forces wide, deep welds — ruling out thin and heat-sensitive parts, and costing the business applications.

During the UltraMAT project, ultrasound was originally being used to disperse material within AA6082-based panels. I proposed applying the ultrasonic excitation during laser welding itself, to investigate whether it could improve weld behaviour — redirecting our part of the project toward ultrasonic-assisted laser welding.

+10%

AVERAGE UTS; +25% BEST TRIAL

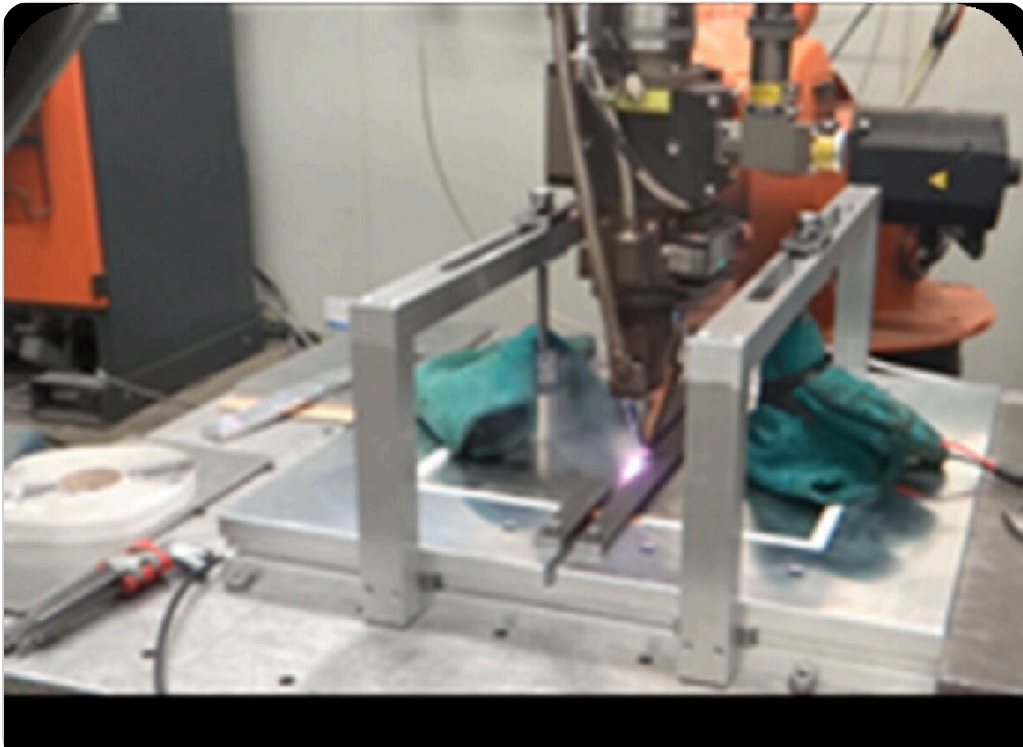
14–62%

STRAIGHTER WELD PROFILE

20–40 kHz

BAND INVESTIGATED

Published in *Metals* and written up for *The Laser User*. It is now the basis of a proposal for high-volume automotive battery pack assembly.



ULTRASONIC-ASSISTED LASER WELDING TRIAL • AA6082

01 CONTEXT

My 2011 MSc thesis found a welding-speed window that let AA6082-T651 solidify without hot cracking. But aluminium's reflectivity meant reaching that window needed very high power density at a fixed beam diameter, and the slow speed produced welds that were wide and deep. Heat-sensitive and thin-walled applications were simply off the table — and every one of those was a job the company could not quote for.

02 THE IDEA

The casting industry has used mechanical vibration for decades to homogenise castings, applied over minutes to hours. Whether it does anything to a solidification event as fast as a laser weld was an open question. The candidate mechanism was ultrasonic grain refinement: vibration impacts, shocks and fluid flow acting as new nuclei and limiting grain growth.

03 METHOD

- Butt welding two 300 × 150 mm AA6082 plates at 1.5, 2 and 4 mm thickness.
- Transducers of varying power clamped to the workpiece, sonicating the plate during welding, melting and solidification.
- Ultrasonic frequency swept across 20–40 kHz to isolate the effective band.
- Every trial run against a no-ultrasound baseline at identical laser settings.

04 EVIDENCE GATHERED

- Surface examination of the weld bead for microcracks and oxidation.
- Cross-section macros measuring weld depth, surface width and root width (aspect ratio).
- Reflected-light microscopy at ×200 for grain structure and dendritic growth.
- SEM at 4000× (BSD, 20 kV) for intermetallic morphology.
- Tensile testing to failure against the baseline.

05 THE EVIDENCE, SIDE BY SIDE

Left of the handle is the conventional laser weld. Right is the same weld with ultrasound applied. Same material, same laser settings.



FIG. 1 – WELD SURFACE

Microcracks along the centreline of the conventional weld, which act as crack initiation zones, disappear completely when ultrasound is used. The plasma cloud is visibly reduced too — the ultrasonic waves stabilise the keyhole, so vaporised metal escapes without a pressure build-up.

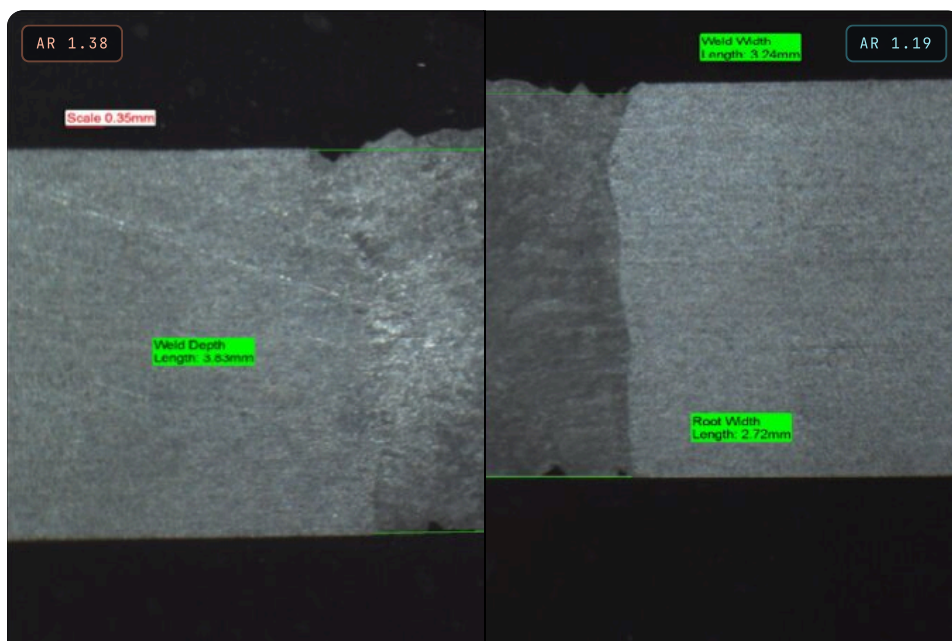


FIG. 2 – CROSS-SECTION & ASPECT RATIO

Aspect ratio is surface width divided by root width; closer to 1.0 means a straighter, more parallel-sided weld. Ultrasound took it from 1.38 to 1.19 on this coupon. Across the study the improvement ranged from 14% to 62%, with 40 kHz best on thin plate and 28 kHz best at 4 mm.

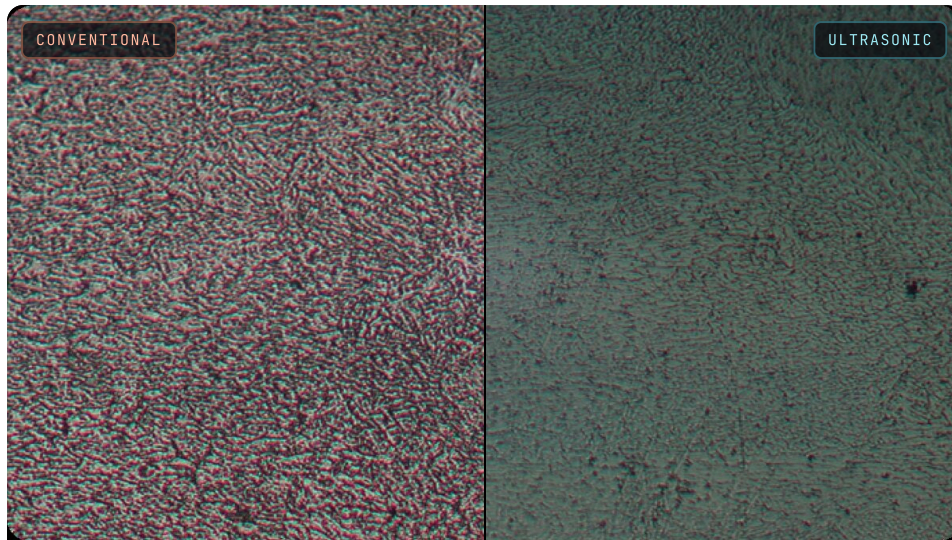


FIG. 3 – GRAIN STRUCTURE, *200

The conventional weld shows dendritic growth in several locations and an unevenly distributed grain structure, with grains crowding around the dendrites. With ultrasound there are no agglomerations, the structure is even, and the only denser region is where two solidification fronts meet.

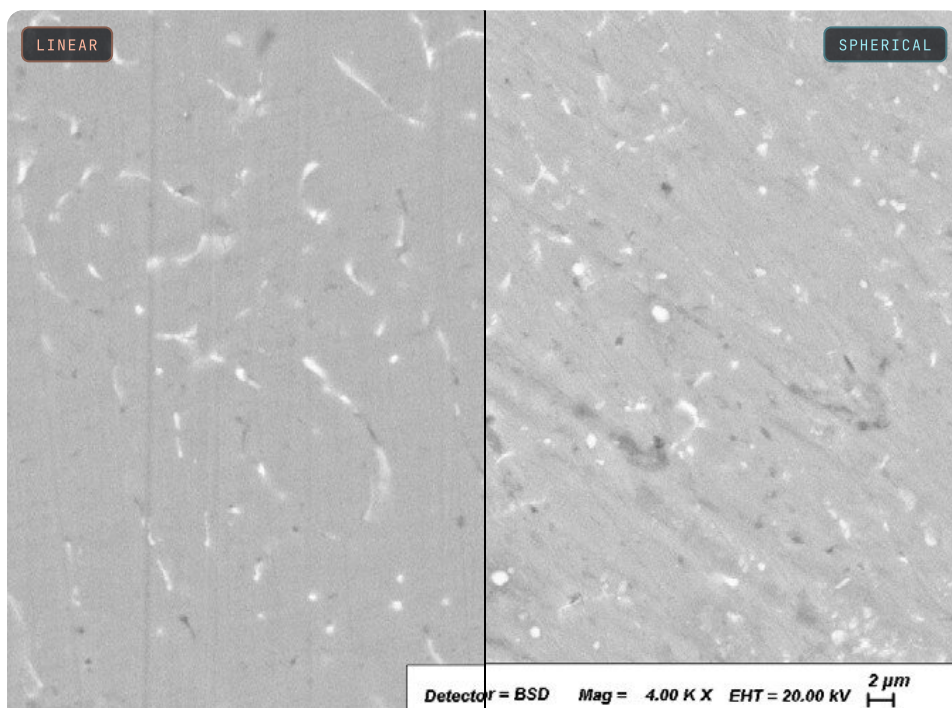


FIG. 4 – INTERMETALLICS, SEM 4000× (BSD, 20 KV)

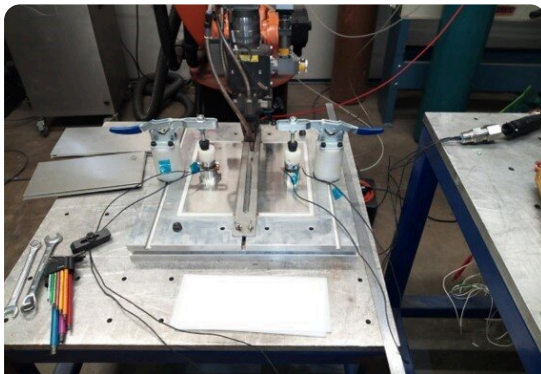
This is the finding that matters most. Without ultrasound the intermetallic compounds form long linear structures — textbook crack initiators. With ultrasound they are smaller and largely spherical, which resist crack initiation far better. That change in morphology accompanied the measured strength increase.

06 RESULTS

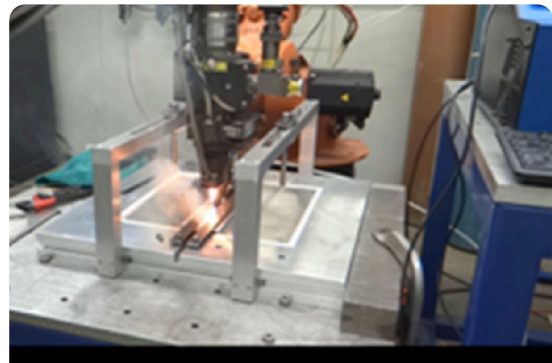
MEASURE	CONVENTIONAL	ULTRASONIC- ASSISTED	CHANGE
Ultimate tensile strength — study average	baseline	+10%	▲ consistent across trials
Ultimate tensile strength — best trial	baseline	+25%	▲ 80 W, 28 kHz, 25 mm/s
Weld profile straightness (aspect ratio)	AR 1.38	AR 1.19	▲ 14–62% across thicknesses
Centreline microcracking	present	eliminated	▲ in every sonicated trial
Intermetallic morphology	linear	spherical	▲ fewer crack initiators
Plasma cloud & spatter	baseline	reduced	▲ improved keyhole stability

Cavitation was identified as a likely significant contributor. Collapsing bubbles generate local forces strong enough to break chemical bonds and raise local temperatures towards 10,000 K; bubble growth and collapse take roughly 1 ms and 150 μ s respectively, so those forces can act even under a rapidly moving laser spot. The calculated cavitation threshold was exceeded at the antinodes. Further modelling was proposed to confirm the role it plays.

07 THE RIG AND THE PROCESS

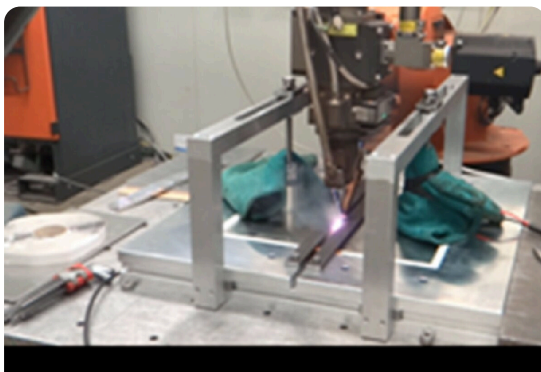


Transducers bonded to the plate, under a robot-mounted laser head.



BASLINE

Welding without ultrasound — larger plasma plume.



WITH ULTRASOUND

Stabilised keyhole, reduced plasma and spatter.

Peer-reviewed and published as *Improvements in the Microstructure and Mechanical Properties of Aluminium Alloys Using Ultrasonic-Assisted Laser Welding* in *Metals* 12(6):1041 (2022), and written up for the AILU membership in *The Laser User*, Issue 106. I presented the concept at Ultramat open days in 2019 and 2020, and took it to ILAS in March 2023 as *Soni-Laser* — a route to high-volume assembly of automotive battery packs. Both UltraMAT (2017) and Soni-Laser (2022) were Innovate UK-funded collaborative projects.

Brunel Innovation Centre provided the ultrasonic expertise and equipment. At CWT I led the welding work, trial planning and macro analysis; tensile testing was carried out independently by TISICS.

Read the paper (<https://doi.org/10.3390/met12061041>)

Case 02 **FEA** AUTOMOTIVE 25CRM04 / AISI 4130

Stopping transverse cracking in a segmented clutch-drum weld

A customer's clutch drum for an armoured executive vehicle had to survive a minimum of 100,000 miles under extreme conditions at a safety factor of 3.0. The first five weld segments went in cleanly. The next five cracked. I was asked to determine the weld profile that would actually hold — and why the existing one was failing.

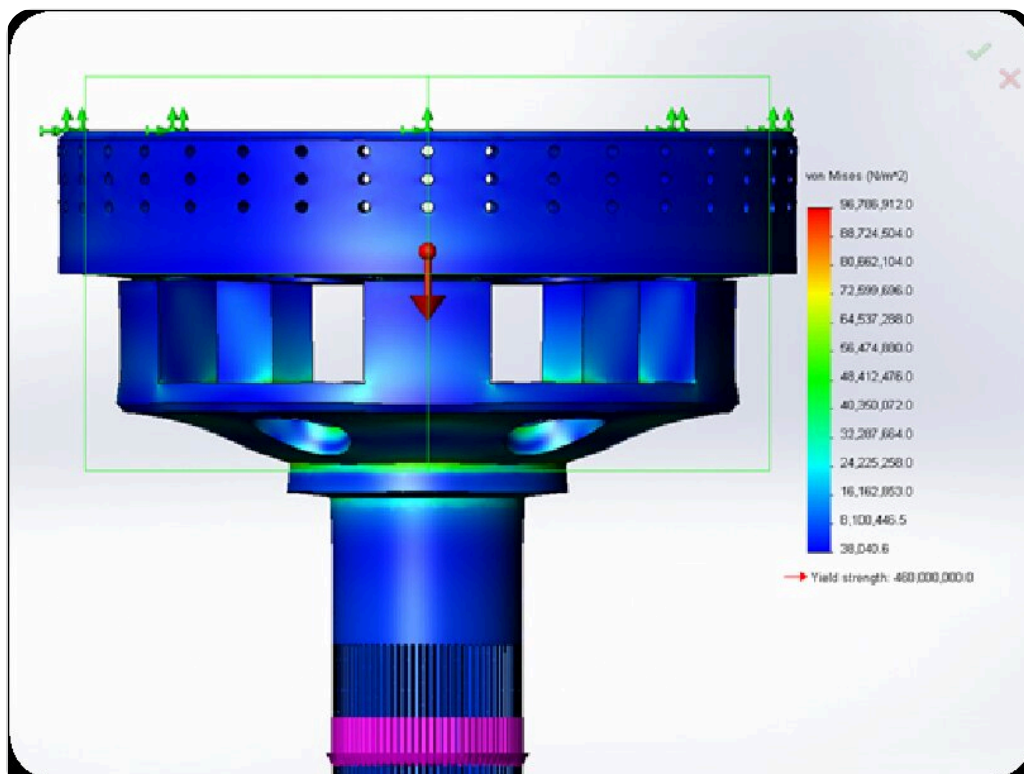
3.0

SAFETY FACTOR REQUIRED

100k

MILES SERVICE LIFE

The corrected process ended the cracking, met the customer's requirement, and brought in an influx of similar work from them and their partners.



FEA TRACTION STUDY • VON MISES STRESS

01 THE PROBLEM

The top surface carried two distinct weld zones — five segments on an outer diameter and five on an inner. In theory that is unremarkable, and the first five went in without issue. The second five produced **transverse cracking** in the earlier welds: cracks running perpendicular to the weld direction, the signature of longitudinal residual stress. The initial welds also showed material contracting away from the centreline, the mark of low-melting-point constituents such as iron sulphites.

02 WHAT HAD TO BE ESTABLISHED

- The **minimum permissible weld depth** — weld size drives residual stress, so oversizing the weld was itself part of the problem.
- The **weld sequence** that minimises accumulated thermal stress across ten segments.
- The **laser parameters** that produce a weld profile able to resist cooling stresses.

03 MODEL & ASSUMPTIONS

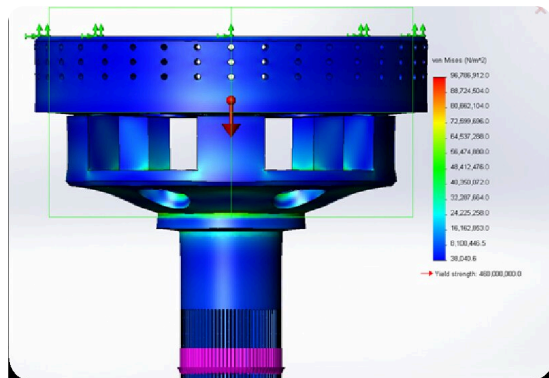
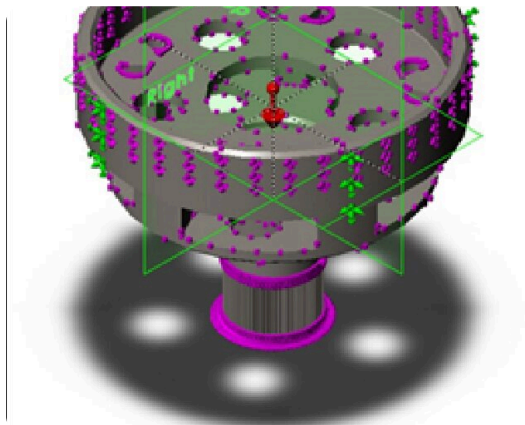
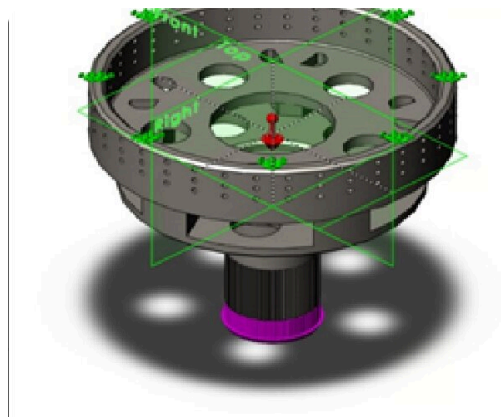
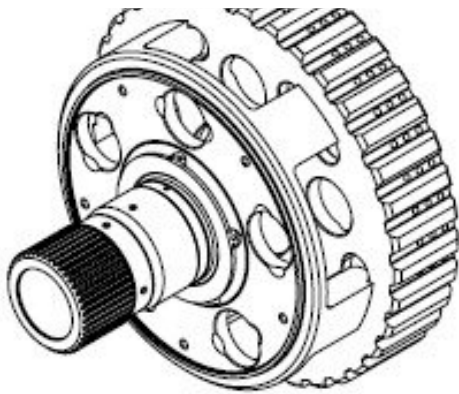
Material	25CrMo4 (AISI 4130)
Geometry	Assembly modelled in SolidWorks as a solid , simulating a fully welded joint
Inner weld	3 mm solid
Traction study	Top surface of the drum edge pinned; 5,000 N applied to the shaft to simulate shear on the welds
Torque study	Drum side surface pinned; 500 N·m applied to the shaft
Solvers	Cosmos and Ansys

04 OUTCOME

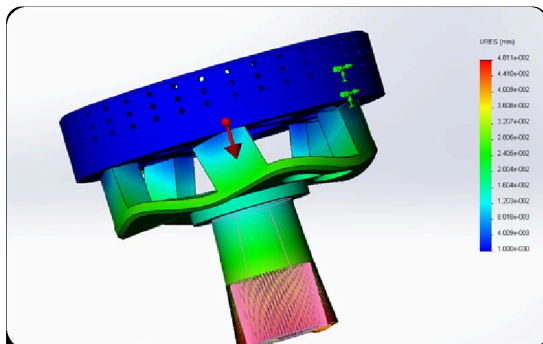
The traction and torque analyses located the zones prone to the highest stress and established that a **3 mm weld depth** was sufficient for the load case. That answered the structural question: the weld did not need to be larger.

The cracking was a separate, metallurgical problem. A distinct process investigation into weld sequence, heat input and the low-melting-point constituents in the parent material produced the revised welding approach that stopped it.

Field testing by the customer exceeded their expectations. The assembly has been in service since **2015**, and the work led directly to more of the same from that customer and their supply chain partners.

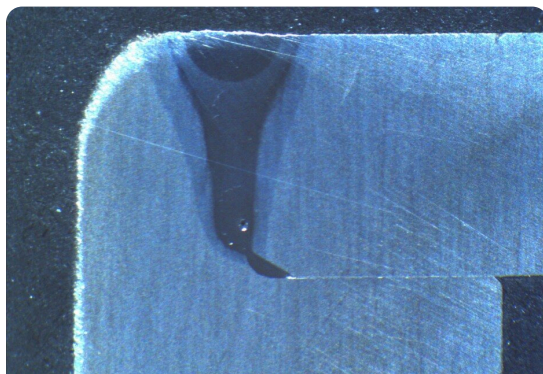
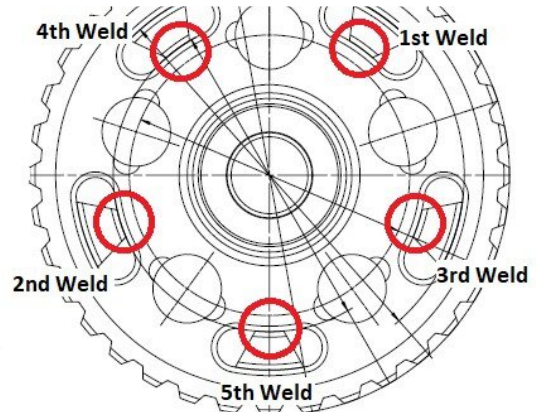


von Mises distribution under 5,000 N.



TORQUE RESULT

Confirmed 3 mm weld depth was adequate.



Macro of a representative weld coupon.

Diagnosing a decline, then rebuilding the process behind it

A large two-year production contract had pulled the company's focus away from production and into tooling repair — a reactive market that is hard to plan around — and nothing had been put in place to replace the revenue when it ended. FY2017 is the low point in the ledger. That year the company went through a management restructure: Alistair Houghton became General Manager, and I took on more of the technical, quality and business-improvement work. We addressed the operational causes together.

~£600k

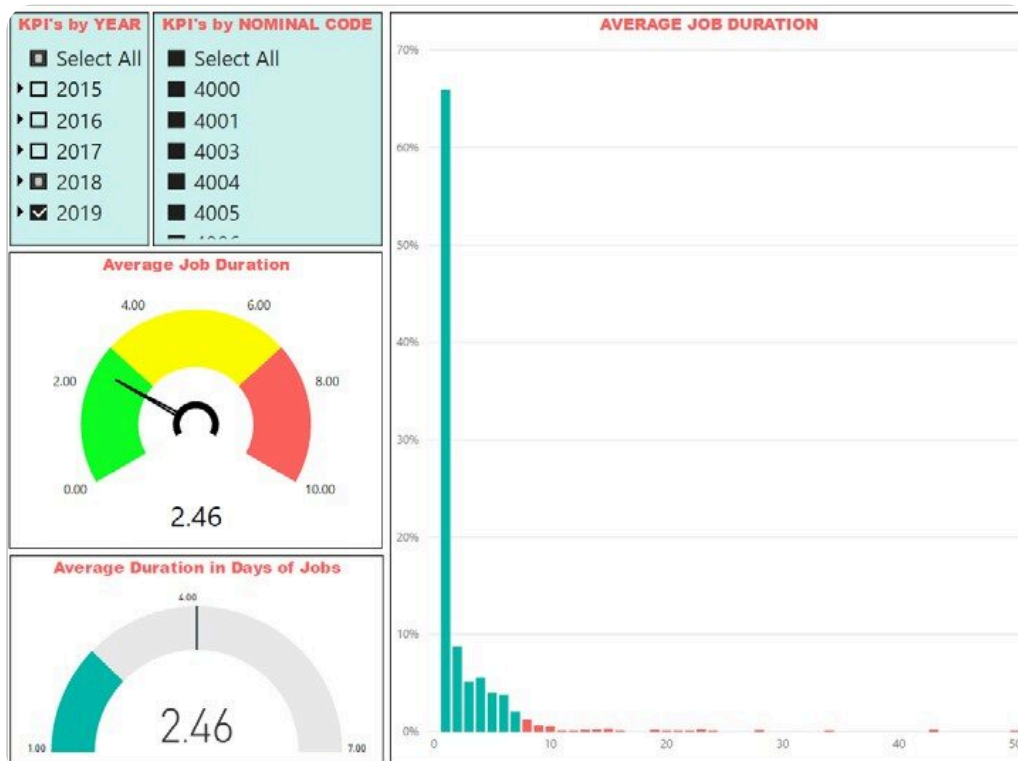
FY2017, THE LOW POINT

~£1.84m

FY2026

Internal Sage ledger, nominal 4000–4999, identified non-trading income excluded. Statutory turnover may differ because of year-end journals.

Turnaround time, non-conformances and monthly revenue. Fix those three and the rest followed. Turnover recovered from the FY2017 low, and the management team went on to own the company.



KPI DASHBOARD • POWER BI

01 THE COMMERCIAL REALITY

A firm supplying a niche service like laser welding survives on two things: it has to add real value to the goods passing through it, and it has to hold a healthy customer base. Added value depends on the workforce and the technology. The customer base depends on part quality, turnaround time, and whether customers are willing to recommend you.

In 2012 a large two-year production job shifted the whole business's focus. When it foreseeably ended, there was no plan to replace it, so the company drifted into tooling repairs — reactive work that is extremely hard to plan around. Less capital came in, profitability fell, and the trend was clearly unsustainable.

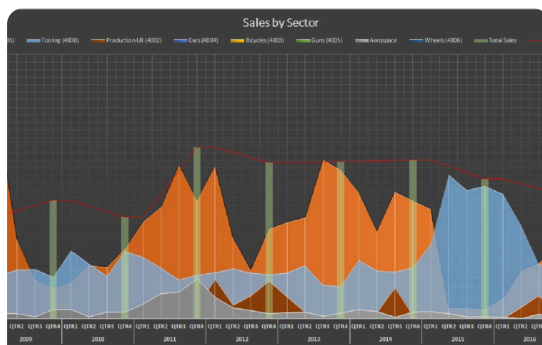
02 INSTRUMENTING THE BUSINESS

Before proposing anything I built the measurements. I designed a set of KPIs to establish the current state and the trend, using Excel for the analysis and Power BI for the data manipulation and dashboards. Three measures turned out to carry the signal:

- **Turnaround time per application**
- **Non-conformances raised**
- **Monthly revenue**, segmented by sector

03 WHAT THE DATA SAID

Turnaround time per application was excessively long, and the non-conformance count was high. Together those two accounted for much of the customer attrition — customers were waiting too long and receiving parts they had to question. Neither problem needed capital expenditure to fix. They needed process.



SALES BY SECTOR

The shape of the problem, over time.



Average job duration by year and code.



REVENUE VS TARGET

Monthly tracking of the recovery.

04 WHAT I CHANGED

- Wrote formal **working procedures** for daily production and welding activity.
- Introduced additional **inbound and outbound quality control**.
- Ran **staff training built around a quality culture** — stopping cross-contamination, and setting an explicit zero-defect target rather than an acceptable rejection rate.
- Shifted the business focus back to **production**, now underpinned by a quality culture that genuinely added value to the welding service.

05 RESULTS

- Customer numbers increased, and existing customers began **referring the company to their partners**.
- Turnover **more than tripled** from the FY2017 low by FY2026.
- The multi-year sales decline was **fully reversed**.
I became a **shareholder** in November 2018.
- The recovery held. In 2025 I completed a **management buyout** of the company alongside my two fellow directors — the strategy I wrote in 2016 is now the business I co-own.

Case 04

CRAFT

SINCE 2014

MARKING & ENGRAVING

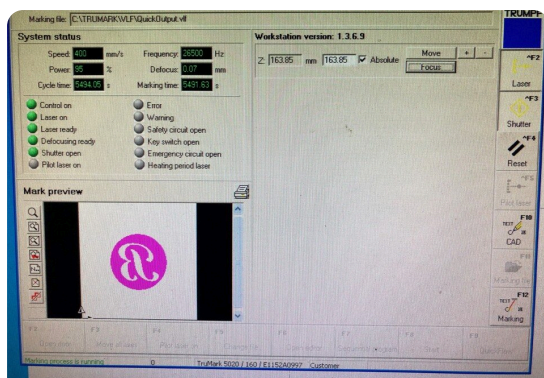
Laser marking and deep engraving, programmed and proven

Designing and operating laser markers since 2014 — programming, fixturing, and dialling in depth and contrast across tool steels, aluminium and stainless. Traceability codes, certification marks, branding and deep engraving on production tooling, plus the operator training and process support that keeps it repeatable.

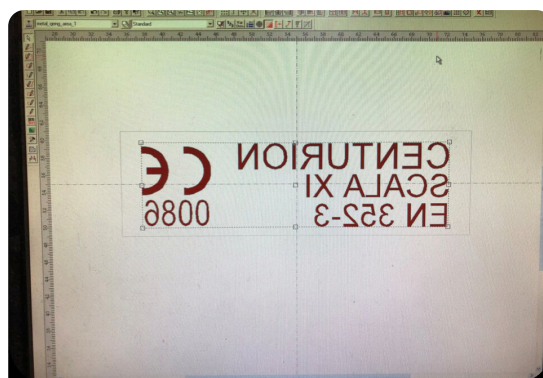


DEEP ENGRAVING • TOOLING INSERT

01 PROGRAMMING THROUGH TO FINISHED PART



Parameters set in TruTops Mark before firing.



Mirrored text for a mould insert.



Clamped and aligned, pilot beam confirming position.



Consistent depth, clean character walls.



Circular component being traced in the fixture.



Standard, notified body and part identity.



Legibility through parameters, not depth.

Instrumenting the business, ten years on

Case 03 was built in Excel and Power BI in 2016. By 2026 the business was three times the size, the auditor's report named the KPI spreadsheets by hand, and every number still ran through me. So I replaced them with an application the company could run without me: a desktop KPI dashboard that imports directly from the accounts ledger, measures the three things that matter, and forecasts the year ahead.

85%

JOBS TURNED ROUND WITHIN FIVE DAYS, AGAINST A TARGET THAT DID NOT EXIST BEFORE

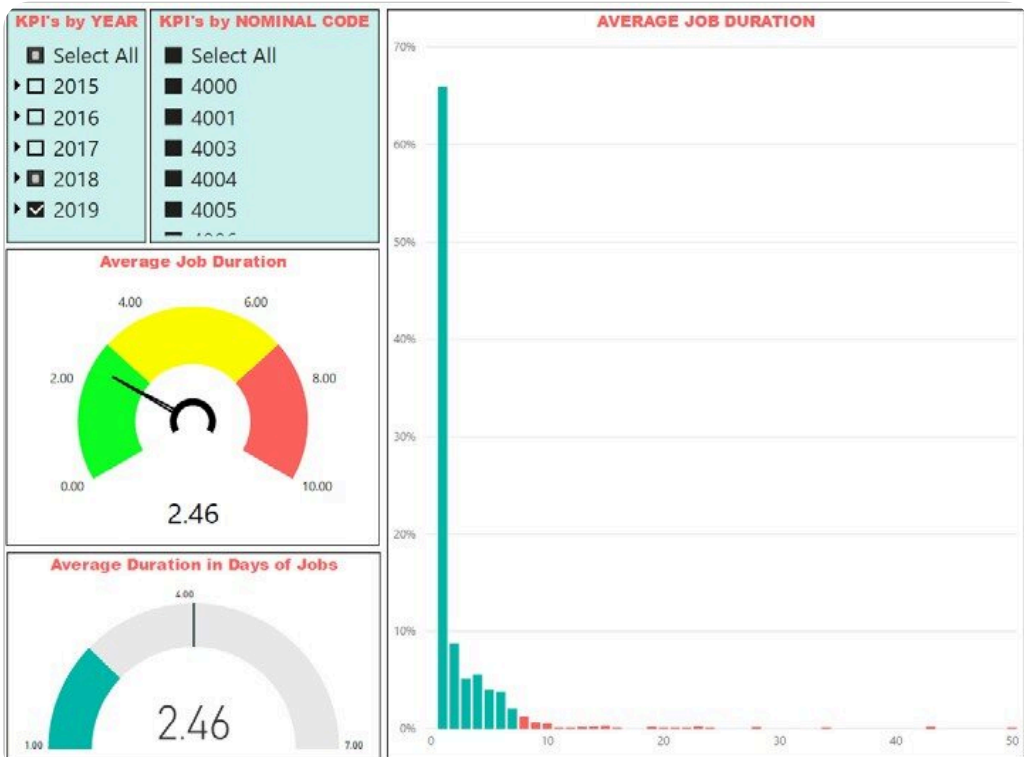
±1.5%

FORECAST ACCURACY IN STABLE YEARS, ROLLING-ORIGIN BACKTEST

-19%

ENERGY PER UNIT OF SALES, YEAR ON YEAR (ISO 14001)

Built for the December 2026 surveillance audit. Management review now runs on it.



PREDECESSOR DASHBOARD, 2016 – REPLACED BY THE APPLICATION DESCRIBED HERE

01

WHAT IT REPLACED

Three Excel pivot dashboards — sales analysis, job completion and non-conformance analysis — maintained by hand and cited by name in the auditor's December 2025 report. Twenty-six years of invoice history and eleven years of nominal-ledger data, imported from Sage exports.

02 WHAT IT MEASURES

Eleven views: sales against target, turnaround SLA, profit and loss, non-conformances with effectiveness review, energy intensity, calibration register, forecast, reports, imports, settings, home.

Turnaround now reports the share of jobs closed within five working days against a formal 85% target. The earlier measure had been an average of jobs falling between three and five days, with no target at all.

Setting the target required documenting which orders count — framework and paused orders are excluded, and the exclusion is recorded.

03 FORECASTING

Holt-Winters exponential smoothing with additive seasonality and damped trend, trained on monthly invoiced sales since 2010. Validated by a three-year rolling-origin backtest: projected annual totals landed within $\pm 1.5\%$ of actuals in the two stable years. The step-change year was under-forecast by 28% — which the method reports rather than hides.

04 ENGINEERING

Python, PyQt6 and SQLite; packaged with PyInstaller and an Inno Setup installer in two flavours, Writer and read-only Viewer, so colleagues see the same numbers without being able to change them. Two automated validators run as the release gate. First installed on a second machine in August 2026.

“

His diligent and methodical approach to problem solving has frequently been a calming influence on our workforce when things aren't going to plan.

Written in support of my application for chartered status, the letter describes a cold cracking problem in gear-to-shaft welds that was not occurring for the normal metallurgical reasons. I traced it to the composition of the shaft material, redesigned the joint to use a filler pass with a specific grade of wire chosen to counteract that metallurgy, modelled and proved the change, and presented the solution to the customer — gaining approval throughout the supply chain and saving the project.

Alistair Houghton

General Manager, Carr's Welding Technologies Ltd — February 2020

Written when Alistair was my manager. We went on to buy the company together, and he is now my co-owner — so read it as the view of someone who backed his own judgement with his own money, not as an independent reference.

Published, peer-reviewed, and presented

When the available technology could not do what customers needed, the answer was to research a better one — and then to share it, in journals, trade press and from the front of the room.

PEER-REVIEWED JOURNAL

Improvements in the Microstructure and Mechanical Properties of Aluminium Alloys Using Ultrasonic-Assisted Laser Welding

Teyeb, A.; Silva, J.; Kanfoud, J.; Carr, P.; Gan, T.-H.; Balachandran, W.
Metals 2022, 12(6), 1041. MDPI.

Cited by **23** subsequent works, including follow-on research at the Brunel Innovation Centre extending the method to dissimilar-metal joints.

Crossref, September 2026

Research funded by Innovate UK (Ultramat, 2017; Soni-Laser, 2022).

10.3390/met12061041 (<https://doi.org/10.3390/met12061041>) [OPEN ACCESS](#) · CC BY 4.0

TRADE PUBLICATION

Improving Al Alloy Properties Using Ultrasonic Laser Welding

Silva, J. & Teyeb, A.

The Laser User, Issue 106, Autumn 2022 — the journal of the Association of Industrial Laser Users (AILU).

INVITED FEATURE

MSC THESIS

Laser Welding of Aluminium Rings - Autogeneous Welding of Aluminium Alloy AA6082-T651

Silva, J.

Instituto Superior Técnico, Universidade Técnica de Lisboa. Published 31 May 2011. Experimental work carried out at Carr's Welding Technologies Ltd.

AUTOGENOUS WELDING OF AA6082-T651 WITHOUT FILLER OR HEAT TREATMENT

PODCAST

TWI Innovation Network Podcast, Episode 4

19 March 2021. In conversation with Dr Cem Selcuk, Head of Business Development at the TWI Innovation Network, alongside Phil Carr.

On the complexities of laser welding, and where artificial intelligence and collaborative robots could take the industry.

Listen (<https://www.twi-global.com/innovation-network/media-and-events/press-releases/2021/the-twiin-podcast-with-carrs-welding>)

GUEST

Invited talks & presentations

MAR 2023	Soni-Laser — ultrasonic-assisted laser welding for high-volume assembly of automotive battery packs	ILAS
MAY 2022	Design for laser welding	Carr's Medical Industry Webinar
FEB 2021	New technologies and advancements in the welding industry	ILAS
JAN 2020	Ultrasonic-assisted laser welding	Ultramat Open Day
MAR 2019	Guiding hand — from welding concept to production	ILAS
JAN 2019	Ultrasonic-assisted laser welding — the concept	Ultramat Open Day
FEB 2018	Laser welding at Carr's Welding Technologies	Jaguar Land Rover
DEC 2017	Laser welding at Carr's Welding Technologies	TWI
NOV 2017	Laser manufacturing of lightweight structures	AILU Lightweighting Workshop, Warwick
MAY 2017	Carr's Welding — capability overview	Renault
DEC 2015	Laser surface engineering	AILU Workshop
SEP 2015	Successful industrial welding techniques and applications with laser & electron beam	AILU Workshop

From the shop floor to the board table

I first walked into Carr's Welding Technologies in February 2010, to run the welding trials for my thesis. Today I set the company's technical direction and co-own it.

SEP 2021 - PRESENT

Technical Director & Co-owner

CWT Group · Kettering, UK · cwt-group.com (<https://cwt-group.com>)

Shareholder since November 2018; statutory director since September 2021; co-owner from October 2025.

Trading name of Carr's Welding Technologies Ltd (no. 03921135) since the September 2025 rebrand.

TECHNICAL

- Evaluate material weldability at quote stage and develop strategies that minimise welding issues in materials with poor weldability.
- Calculate weld strength at design stage and advise customers on the weld depth and profile their part actually requires.
- Advise customers on the right welding technology for their application, and design new or improved parts to suit.
- Calculate the parameters for every laser operation; assess and procure new technology.
- Guide every R&D project — strategy, parameters and welding method — through to the required outcome.
- CNC technical support and training; laser marking programming, support and training.

QUALITY

- Produce the production and welding procedures used in daily work.
- Run internal audits and maintain the quality management system to ISO 9001:2015, with controls kept beyond the minimum the standard requires.
- Carry out all quality inspections: visual, dye penetrant, helium leak, macro and micro analysis.
- Issue certificates of conformance, inspection reports and first article inspection reports.
- Improve process quality using PFMEA and PPAP for automotive and aerospace customers.
- Build and maintain the company's stock control and KPI calculation software.
- Identify non-conformances and develop corrective actions that stop them recurring.

KEY ACHIEVEMENTS

Developed a program that calculates welding parameters from customer requirements. Worked with the General Manager to change the profile and strategy of the company's business plan and carry the workforce with it, as turnover recovered from the FY2017 low. Responsible for the content and redesign of the company website.

APR 2013 — AUG 2021

Quality & Technical Manager

Carr's Welding Technologies Ltd · Kettering, UK — now trading as CWT Group

SCOPE

- All quality and technical assurance across a business serving aerospace, automotive, pharmaceutical, Formula 1, oil & gas, naval and tool engineering.
- Created and maintained the KPIs the business is still measured on.
- Improved daily workflow using PERT, Critical Path Method and Gantt planning.

ALSO

- Coached the workforce on best welding practice.
- Created key company documents — employee handbook, contracts, customer registration forms, training matrix.
- Created and implemented the ISO 14001 environmental management system in 2021.

RECOGNITION

Became a shareholder in November 2018, after the 2017 management restructure and the recovery work that followed — the first step towards co-ownership in 2025.

JUN 2014 — NOV 2018

Quality & Technical Manager — CONCURRENT

Carr's Welding Products Ltd · Kettering, UK — aerospace & automotive (sister company; dissolved 2019)

RESPONSIBILITIES

- In charge of all quality and technical assurance for the sister company.
- Responsible for elevating its quality management system to AS9100.
- Planned strategies to improve production workflows; created and maintained KPIs.

ACHIEVEMENT

- Created new ways of working by planning production workflows and applying project management tools — including the spreadsheet systems that made the quality management system uplift possible.

MAR 2012 — MAR 2013

Technical & Quality Engineer

Carr's Welding Technologies Ltd · Kettering, UK

HOW IT STARTED

- Recruited on graduation, by the owner of the company whose equipment I had just spent my master's research on.
- Brought in specifically to raise the company's quality level — the brief that shaped everything that followed.

WHERE IT LED

- Promoted to Quality & Technical Manager within a year.
- This is where my career as a mechanical engineer began, and where the pursuit of a meaningful impact on the laser welding industry started.

FEB — MAR 2010

MSc thesis welding trials

Carr's Welding Technologies Ltd · Kettering, UK

WHERE IT ALL BEGAN

- Came over from Instituto Superior Técnico to run the practical trials for my MSc thesis, begun in Lisbon the previous September: laser welding of aluminium alloy AA6082 — the crack-prone alloy I have been chasing ever since.
- Those trials became my thesis, then my first job, then the ultrasonic research, and eventually the company itself.

Qualifications, standards and toolset

Formal education in Lisbon, standards implemented in Kettering, and the software I actually use week to week.

EDUCATION

MSc Mechanical Engineering

Instituto Superior Técnico, Universidade Técnica de Lisboa

MAR 2007 — JAN 2012

Thesis: *Laser Welding of Aluminium Rings - Autogeneous Welding of Aluminium Alloy AA6082-T651*, published 31 May 2011; experimental work carried out at Carr's Welding Technologies Ltd. Investigated porosity and hot cracking in AA6082-T651 and demonstrated viable autogenous welding without filler or pre/post-weld heat treatment. Highest grade in the year for Heat Transfer.

BEng Mechanical Engineering

Instituto Superior Técnico, Universidade Técnica de Lisboa

SEP 2003 — SEP 2007

Honourable mention in the SolidWorks 2004 International Contest for the project "Monster Truck".

Certificate of Proficiency in English (CPE)

Cambridge School Learning Institute, Lisbon

SEP 2003 — SEP 2004

Level C2 on the Council of Europe Common European Framework; Level 3 on the UK National Qualifications Framework.

STANDARDS & ASSURANCE

- ✓ **AS9100D** — wrote the complete QMS for Carr's Welding Products, an aerospace welding business, prepared the assessments and took the system through to certification. That company closed in 2018; many of its controls were carried into CWT, which is certified to ISO 9001, not AS9100.
- ✓ **ISO 9001:2015** — quality management. Implemented, maintained, internally audited.
- ✓ **ISO 14001** — environmental management. Created and implemented in 2021.
- ✓ **Recertification, December 2025** — ISO 9001:2015 and ISO 14001:2015, two-day external audit: zero major, zero minor non-conformances, one opportunity for improvement.
- ✓ **Welding procedure standards** — WPS authored to ISO 15609-4; weld quality levels to EN ISO 13919-1 and -2; procedures qualified by customer examination and approval under EN ISO 15611; visual inspection to EN ISO 17637.
- ✓ **Records** — quality records retained 10 years for standard customers and 30+ years for aerospace; paper originals are the retained record.
- ✓ **PFMEA & PPAP** — automotive and aerospace process quality.
- ✓ **FAIR, CofC, IR** — first article inspection reports, certificates of conformance, inspection reports.
- ✓ **Inspection methods** — visual, dye penetrant, helium leak testing, macro and micro analysis.
- ✓ **Project control** — PERT, Critical Path Method, Gantt planning.

CORE COMPETENCIES

Laser welding & process development EXPERT



Weld metallurgy & weldability EXPERT



Quality management systems EXPERT



Laser marking & CNC programming EXPERT



Mechanical design & FEA ADVANCED



Project & production management ADVANCED



Data analysis & KPI reporting ADVANCED



TOOLSET

CAD / CAE / CAM

SolidWorks

Ansys

Cosmos

Autodesk Inventor

AutoCAD

Solid Edge

Mastercam CAD/CAM

Moldflow Plastics Insight

PROGRAMMING

Python

SQL / SQLite

PyQt6

React

JavaScript

VBA

Matlab

Maple

Fortran 90

Java

Saia PG5

HTML / CSS

MACHINE & PROCESS

TruTops Mark

Disk lasers

Fibre lasers

Oscillating (wobble) welding

CNC programming

BUSINESS

Power BI

Excel

Word

Access

Visio

PowerPoint

Outlook

07 - CONTINUOUS DEVELOPMENT

Still learning, deliberately

A formal CPD record, and a decade of conferences, technology assessments and workshops across the UK, Germany, Austria, Ireland and the USA.

Continuous professional development

15 records • 2021-2023

Jan - Apr 2021 • Course

100 Days of Python Bootcamp, Udemy. Learnt to program in Python and built my own portfolio website.

Feb 2021 • Course

Marketing with LinkedIn Masterclass. Content creation, customer visibility and sales.

Mar 2021 • Podcast

TWI Innovation Network Podcast, Episode 4. Shared my view on where AI and cobots could take the welding industry.

Apr 2021 • Work-based

IPG oscillating laser welding training. Operating and programming the newest oscillating laser welding technology.

Aug 2022 • TWI webinar

An introduction to numeric modelling at TWI. Update on modelling techniques and software across engineering applications.

Aug 2022 • TWI webinar

The fatigue performance of welded joints — a refresher. Fatigue effects on welded joints and strategies to minimise them.

Sep 2022 · TWI webinar

Damascus steel in legend and reality. How Damascus steel is made, and how to tell real from fake.

Sep 2022 · NQA webinar

ISO 14001 — back to basics. Additional skills for the journey to carbon neutrality.

Oct 2022 · TWI webinar

Back to basics — cracking in ferritic steels. Methods to avoid cracking, and techniques for predicting crack sensitivity.

Nov 2022 · NQA webinar

ISO 27001 — back to basics. Benefits of and route to accredited certification.

Nov 2022 · TWI webinar

Failure analysis — methodology, techniques and case studies. Refreshed methodology and picked up further techniques.

Dec 2022 · TWI webinar

NDT of additively manufactured components. New techniques for inspecting additively manufactured parts.

Jan 2023 · NQA webinar

ISO 9001 — internal auditing. How the internal audit process is managed, and how it lifts the quality system.

May 2023 · Struers webinar

Materialographic preparation of titanium and titanium alloys. Theory and technique for titanium sample preparation.

Jun 2023 · TWI webinar

Back to basics — engineering critical assessment. New concepts and methodologies in ECA.

Conferences, workshops & technology assessments

28 events · 2014–2023

Mar 2023	ILAS — Soni-Laser: ultrasonic-assisted laser welding for high-volume assembly of automotive battery packs	PRESENTER
May 2022	Carr's Medical Industry Webinar — Design for laser welding	PRESENTER
Feb 2021	ILAS — New technologies and advancements in the welding industry	PRESENTER
Jan 2020	Ultramat Open Day — Ultrasonic-assisted laser welding	PRESENTER
Nov 2019	Alpha Laser — laser welding technology assessment, Puchheim, Germany	
Oct 2019	Advanced Engineering Show — NEC	
Mar 2019	ILAS — Guiding hand from welding concept to production	PRESENTER
Jan 2019	Ultramat Open Day — Ultrasonic-assisted laser welding concept	PRESENTER
Nov 2018	AILU Jobshop annual meeting — Bystronic	
Oct 2018	Advanced Engineering Show — NEC	
Feb 2018	Jaguar Land Rover — Laser welding at Carr's Welding Technologies	PRESENTER
Dec 2017	TWI — Laser welding at Carr's Welding Technologies	PRESENTER
Nov 2017	AILU Lightweighting Workshop — laser manufacturing of lightweight structures, Warwick University	PRESENTER
Oct 2017	MedTech Medical Technology — Galway, Ireland	
Oct 2017	AILU Jobshop annual meeting — BOC	
Jul 2017	Advanced Engineering Show — NEC	
May 2017	Renault — Carr's Welding capability overview	PRESENTER
Mar 2017	ILAS 2017 — the 5th Industrial Laser Applications Symposium	
Nov 2016	Advanced Engineering Show — NEC	

May 2016	TRUMPF — laser welding technology assessment, Stuttgart, Germany	
Feb 2016	IPG Lasers — oscillating laser welding workshop and trials, Boston, USA	
Dec 2015	AILU — Laser Surface Engineering Workshop	PRESENTER
Sep 2015	AILU Workshop — successful industrial welding techniques and applications with laser & electron beam	PRESENTER
Jul 2015	TRUMPF — laser welding technology assessment, Stuttgart, Germany	
Sep 2014	Magna Steyr — laser setup, training and process approval, Austria	
Jul 2014	TRUMPF — laser welding technology assessment, Stuttgart, Germany	
May 2014	Magna Steyr — laser setup and training, Austria	
Feb 2014	TRUMPF — laser welding technology assessment, Stuttgart, Germany	

Milestones

2003 – 2026

2003	Portuguese National Triathlon Champion
2003	Certificate of Proficiency in English, Cambridge School
2004	Honourable mention, SolidWorks International Contest — project “Monster Truck”
2007	BEng Mechanical Engineering awarded
2010	MSc thesis welding trials at Carr's Welding Technologies (February–March) — laser welding of AA6082
2011	Highest grade of the year in Heat Transfer
2011	MSc thesis on laser welding of AA6082-T651 aluminium rings published (31 May)
2012	MSc Mechanical Engineering awarded (January)
2012	Joined Carr's Welding Technologies Ltd as Technical & Quality Engineer
2013	Promoted to Quality / Technical Manager
2013	Built a program that calculates laser settings from customer requirements
2013	Began compiling the AS9100 QMS for Carr's Welding Products Ltd, formed the following year
2017	Management restructure; began the operational and quality work behind the recovery
2018	Became a shareholder in Carr's Welding Technologies Ltd (November)
2019	Proposed and led CWT's ultrasonic-assisted laser welding trials on the UltraMAT project
2021	Created and implemented the ISO 14001 EMS for Carr's Welding Technologies Ltd
2021	Promoted to Technical Director
2022	Published in <i>Metals</i> — ultrasonic-assisted laser welding of aluminium alloys
2022	Feature article published in <i>The Laser User</i> , Issue 106
2025	Company rebranded as CWT Group (September)
2025	Became co-owner alongside two fellow directors (October)
2025	Recertified to ISO 9001:2015 and ISO 14001:2015 with zero non-conformances (December)
2026	KPI dashboard application deployed, replacing the 2016 spreadsheets
2026	WPS Database under version control with Architecture Decision Records

Got a part that won't weld?

Whether it is a weldability question on a drawing, a joint that keeps failing, a quality system that needs to reach AS9100D, or a research collaboration — I would be glad to talk it through.



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BASED IN

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Professional references available on request.